PROBLEM SET 2

Codes

1. Express the following decimal numbers in (i) 8421 BCD code, (ii) XS-3 code, (iii) 2421 code, (iv) 5211 code

(a) 286 (b) 807 (c) 429.5 (d) 158.7

2. Express the following 8421 BCD numbers as decimals.

(a) 1000 0100 0110 (b) 0110 0111 1001. 1000
(c) 0110 1001 0111. 0101 (d) 1000 0110 0000. 1000

3. Express the following XS-3 numbers as decimals.

(a) 1010 1100 0110 (b) 0101 1011 0111. 1001
(c) 1001 1100 1010. 0111 (d) 0111 0100 1011. 1000

4. Add the following in (i) BCD and (ii) XS-3 codes.

(a) 275 + 493 (b) 109 + 778 (c) 88.7 + 265.8 (d) 204.6 + 185.56

5. Subtract the following in (i) BCD and (ii) XS-3 codes.

(a) 920 – 356 (b) 79 – 27 (c) 476.7 – 258.9 (d) 634.6 – 328.7

6. Convert the following decimal numbers to Gray code.

(a) 8 (b) 23 (c) 96 ` (d) 246

7. Convert the following binary numbers to Gray code.

(a) 1011 (b) 101010110101 (c) 110110010 (d) 100001

8. Convert the following Gray code to binary.

(a) 1111 (b) 101110 (c) 100010110 (d) 11100111

9. Convert the following numbers into Gray code numbers.

(a) 96_{10} (b) 45_{16} (c) 234_{8} (d) 85_{12}

10. Detect and correct the errors, if any, in the even parity Hamming code words and write the correct code

(a) 1100110 (b) 0011101 (c) 0111110 (d) 1010111

11. (a) Given the 8-bit data word 10111001, generate the 12- bit composite word for the Hamming code that corrects and detects single errors.

(b) A 12-bit Hamming code word containing 8 bits of data and 4 parity bits is read from memory. What is the original 8-bit word if the 12-bit read out is as follows?

(i) 1100 1010 0110
(ii) 0011 0110 0101
(iii) 1010 1001 1101